

TASK 2 — Body Measurement Logic

Overview

Body measurement using AI is based on detecting human body landmarks through pose detection systems. These landmarks represent key joints of the body, and distances between them can be used to estimate body proportions such as shoulder width, height, and waist.

The measurements obtained are relative (pixel-based) and can be scaled to real-world units using calibration techniques.

1. Shoulder Width Calculation

Concept

Shoulder width is calculated by measuring the distance between the left and right shoulder landmarks detected by the pose model.

Steps

- Detect left shoulder landmark
- Detect right shoulder landmark
- Extract (x, y) coordinates
- Compute distance using Euclidean formula

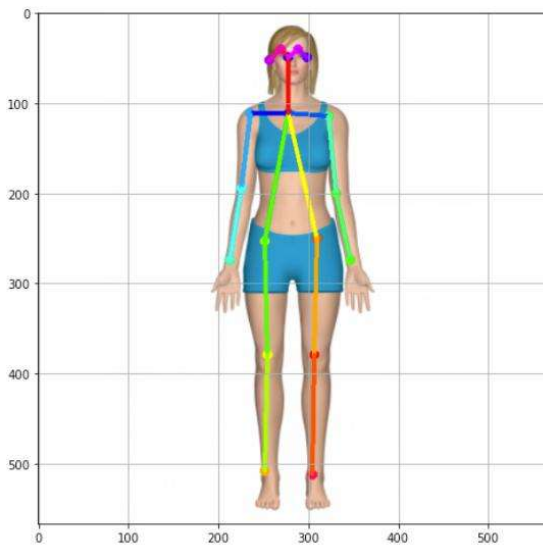
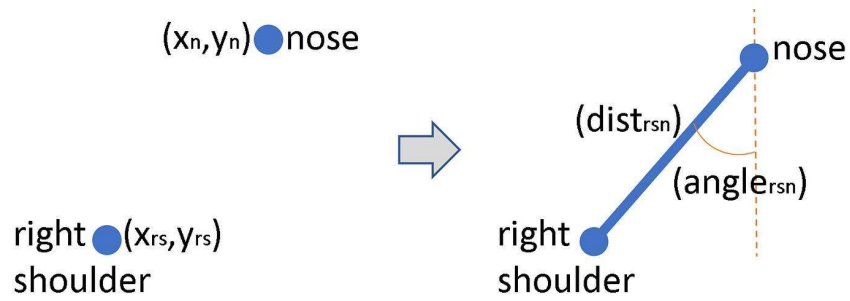
Formula

$$\text{distance} = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Use Cases

- Clothing size estimation
- Body proportion analysis
- Fitness posture tracking

2. Height Estimation



Concept

Height is estimated using the vertical distance between the head (nose landmark) and the ankle.

Steps

- Detect nose (top point)
- Detect ankle (bottom point)
- Calculate vertical distance

Note

- This gives approximate height
- Accuracy depends on full body visibility

Applications

- Virtual fitting systems
- Fitness analysis
- Health monitoring

3. Waist Estimation

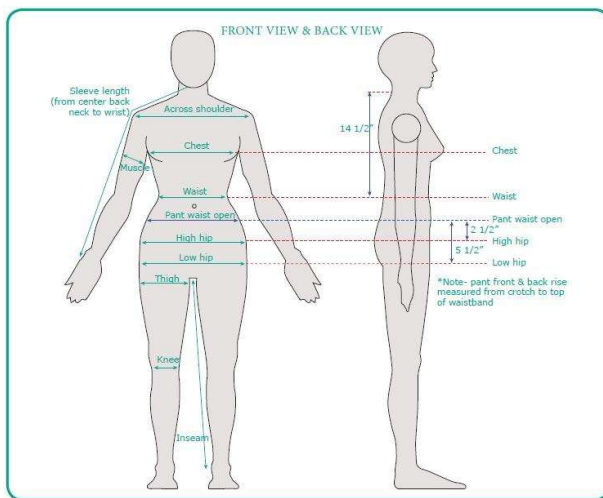
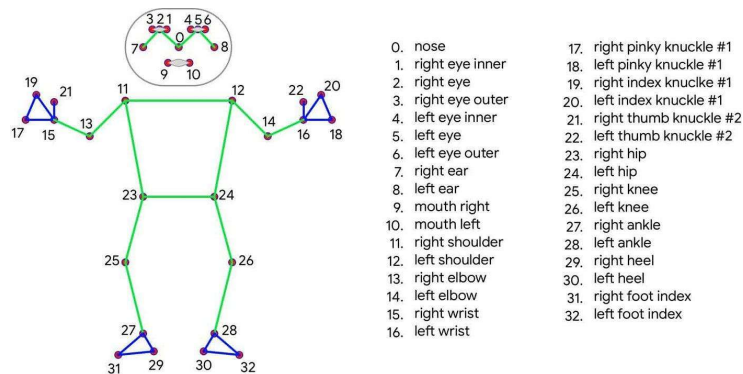


Figure 9.2 Body Measurements

Concept

Waist size is estimated using the distance between left and right hip landmarks.

Steps

- Detect left hip
- Detect right hip
- Compute distance

Applications

- Clothing recommendations
- Body shape analysis
- Virtual try-on systems

Important Considerations

- Measurements are based on pixels, not real units
- Requires calibration for accurate sizing
- Camera angle and distance affect accuracy

Conclusion

Body measurement logic demonstrates how AI can estimate human body proportions using simple geometric calculations on detected landmarks. This forms the foundation for advanced applications like virtual try-on and personalized fashion systems.